
Maintain Integration

Given a use case, identify reporting needs for integration monitoring.

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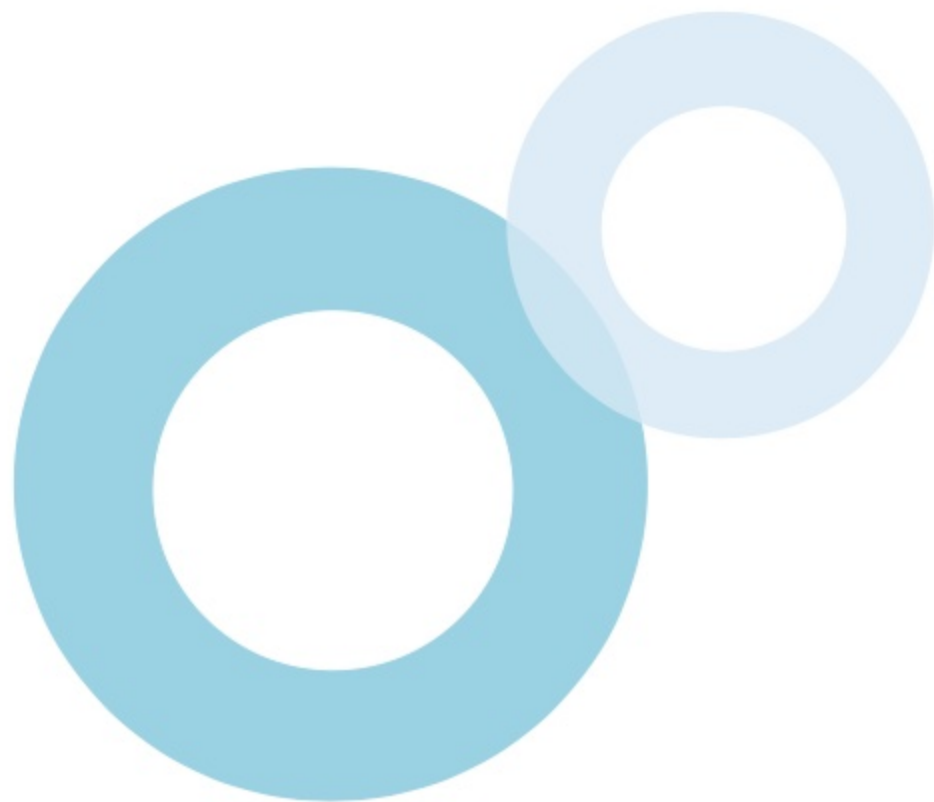
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After studying this topic, you should be able to:

-  Learn more about event monitoring and real-time event monitoring
-  Determine user activities in an integrated environment that can be reported
-  Identify the event objects and their file data schema which are monitored



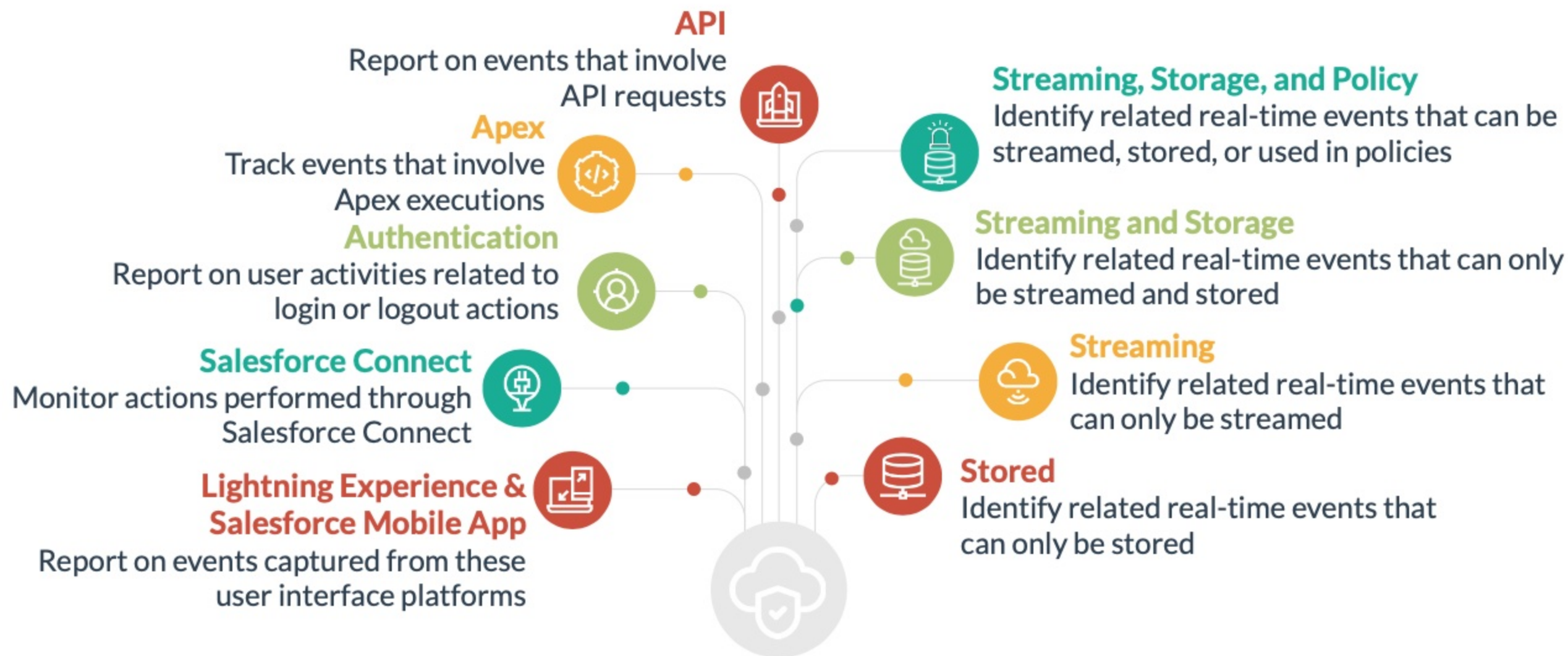
Introduction

This section takes a more in-depth look into **Event Monitoring** and **Real-Time Event Monitoring** to cover other general events that can be captured in an org environment that is integrated with another system. While Event Monitoring store events in **event log files** when resources are available, Real-Time Event Monitoring can **stream** and/or **store** events in near-real time. Furthermore, a **transaction security policy** can be created to intercept a real-time event and determine its next course of action.

The event information at the **object** and **schema** level will be described in order to provide a clearer picture of how granular the captured details are that enable for more **efficient monitoring** and can meet even complex **reporting needs**.



Reporting Needs



Event Monitoring

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Event Monitoring

Event Monitoring is particularly useful for **auditing, compliance, troubleshooting, and performance optimization** in Salesforce environments.

❖ PERMISSION SET

Event Monitoring User permission simplifies access to all Event Monitoring data, including **Event Log Files, Objects, Real-Time Events**, and **Threat Detection**.

❖ SETUP

It is available with **Salesforce Shield** and **Salesforce Event Monitoring** add-ons. Available in Enterprise, Performance, and Unlimited Editions with **Event Monitoring enabled**.

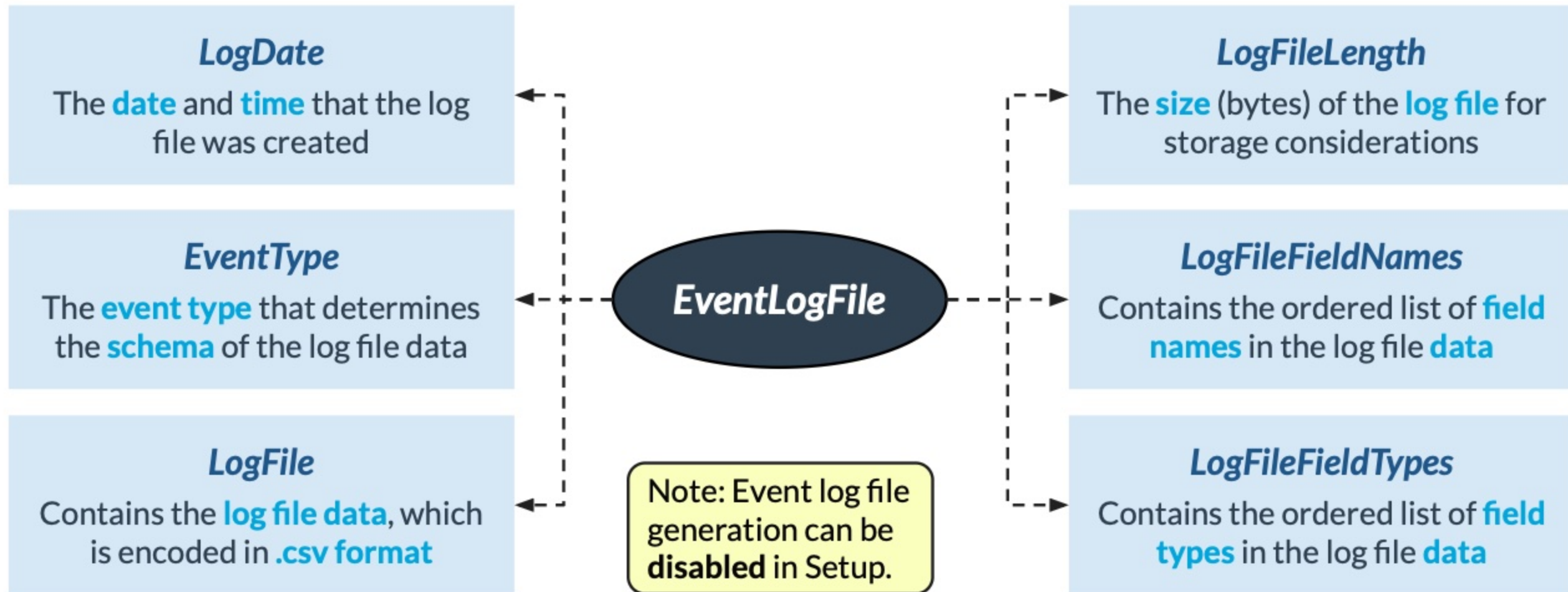
❖ EVENT LOG OBJECT FRAMEWORK

Allow to store event data for the past **30 days** and query any **15-day** window via the API.



The EventLogFile Object

An **event log file** is generated each time an event is captured by the **Event Monitoring** tool. The diagram below provides a closer look on **some** of the fields that is included in the object.



The LogFile Data

The **schema** of a log file data (.csv) depends on the **event type** of the captured event. For example, the following shows the last columns of the log files of an **Apex callout** event versus a **Login** event.

Apex callout event log file data

SIZE	RESPONSE_SIZE	URL	TIMESTAMP_DERIVED	USER_ID_DERIVED	CLIENT_IP	URI_ID_DERIVED
-1	52	"https://api-prod.guidance.sfdc.sh/odat	2022-07-25T09:45:34.435Z	0052v00000YeuUvxxx		

Login event log file data

AUTHENTICATION	TIMESTAMP_DERIVED	USER_ID_DERIVED	CLIENT_IP	URI_ID_DERIVED	LOGIN_STATUS	SOURCE_IP
GCM-SHA384	2022-07-25T09:45:19.441Z	0052v00000YeuUvxxx	130.105.110.xx		LOGIN_NO_ERROR	130.105.110.xx

API Event Types

Event Monitoring can capture events in an integration solution that involve **API requests**. For example, the following **event types** can be used to track the related data.



Apex REST API

Contains details related to a **custom** REST API request



Apex SOAP

Contains details related to a **custom** SOAP API request



API Total Usage

Contains API total usage details related to a **SOAP**, **REST**, or **Bulk API** request



Bulk API

Contains details related to a **BULK API** request



Bulk API 2.0

Contains details related to a **BULK API 2.0** request



REST API

Contains details related to a **standard** REST API request



SOAP API

Contains details related to a **standard** SOAP API request

Example Event Type: REST API Event

When an **inbound REST API call** is captured, the event details are stored in a **log file**. The below describes some of the **fields and data** that are recorded from a captured **REST API event**.

CLIENT_IP

IP address of the client who performed the request

ENTITY_NAME

The **sObjects** that were accessed in the request

METHOD

The **HTTP method** used in the request (GET, POST, PUT, etc.)

REQUEST_STATUS

The **status** of the request such as whether it succeeded or failed

REQUEST_SIZE

The **size** of the **request body** in bytes

RESPONSE_SIZE

The **size** of the **callout response** in bytes.

ROWS_PROCESSED

The **number of rows** that were processed in the request

RUN_TIME

The **total time** spent to complete the request in milliseconds

USER_ID

The ID of the **Salesforce user** who performed the request

Apex Event Types

Event Monitoring can capture events in an integration solution that involve **Apex executions**. The following **event types** can be used to track the related data.



Apex Callout

Contains details related to **external callouts** in an Apex execution



Apex Execution

Contains information about **Apex classes** used in an execution



Apex Trigger

Contains details related to Apex triggers that **were fired**



Apex Unexpected Exception

Contains details of an **unexpected exception** in an Apex execution



Aura Request

Contains details of server-side method calls from **Aura** or **Lightning web components**



Concurrent Long-Running Apex Limit

Details of the concurrent request that **exceeded the limit**

Apex Event Types

Event Monitoring can capture events in an integration solution that involve **Apex executions**. The following **event types** can be used to track the related data.



Continuation Callout Summary

Contains details of the **asynchronous callouts** in an Apex transaction



Named Credential

Contains details of **Apex callouts** that used named credential **endpoints**



Visualforce Request

Contains details of Visualforce **page requests** initiated by users

Example Event Type: Apex Execution Event

Whenever a transaction occurs that involves **Apex classes**, the event details are stored in a **log file**. Here are some of the **fields and data** that are recorded from an **Apex Execution** event.

EXEC_TIME

The total **end-to-end** Apex execution time in milliseconds

IS_LONG_RUNNING_REQUEST

Whether the request is a concurrent long-running request

NUMBER_SOQL_QUERIES

Counts the number of **SOQL queries performed** in a request

QUIDDITY

The **type of execution** (batch, bulk, future, rest, soap, etc.)

URI

The **URI of the page** that received the request

USER_ID

The ID of the **Salesforce user** who performed the request

Authentication Event Types

Event Monitoring can capture events in an integration solution that involve **user login and logout** activities. The following **event types** can be used to track the related data.



Login

Contains details of the **login history** of a user



Logout

Contains details of the **logout actions** of a user

Example Event Type: Login Event

Whenever a user or system **logs into** the org, the event details are stored in a **log file**. The below describes some of the **fields and data** that are recorded from a **Login event**.

API_TYPE

The type of API request **used to login** such as Apex, SOAP, etc.

LOGIN_KEY

A string value used to **associate all events** in a user session

LOGIN_STATUS

Shows **LOGIN_NO_ERROR** if a login attempt was **successful**

SOURCE_IP

The **IP address** where the login request came from

TIMESTAMP

The **time (GMT)** that the Salesforce service was accessed

TLS_PROTOCOL

The TLS protocol **used to login** (1.0, 1.1, or 1.2)

USER_ID

The **Salesforce User ID** associated with the request

USER_NAME

The **Salesforce Username** associated with the request

USER_TYPE

The **category** of the **user license** associated with the user

Salesforce Connect Event Types

Event Monitoring can capture events that are performed through a **Salesforce Connect** integration. The following **event types** can be used to track the related data.



External Cross-Org Callout

Contains details of data callouts made through a **cross-org adapter**



External Custom Apex Callout

Contains details of data callouts made through **custom adapters**



External OData Callout

Contains details of data callouts made through a **OData 2.0/4.0 adapter**

Example Event Type: External OData Callout Event

Whenever a user interacts with an **external object**, the event details are stored in a **log file**. Here are some of the **fields and data** that are recorded from an **External OData Callout event**.

ACTION

Whether the callout action was a **query**, **upsert**, or **delete**

BYTES

Indicates the **size (bytes)** of the **result set**

ENTITY

Name of the **external object** that was **accessed**

FILTER

Field expressions used to filter the records that was returned

ROWS_FETCHED

Number of records returned by the callout

SELECT

A **comma-separated** list of **fields** used in the query

STATUS

Whether the callout action to the **external system** was successful

THROUGHPUT

The **number of records** that were retrieved in **one second**

TOTAL_MS

Total time taken to **query and retrieve** the result set

Lightning Experience and Salesforce Mobile App Event Types

Event Monitoring can capture user-initiated events in **Lightning Experience** or the **Salesforce mobile app**. The following **event types** can be used to track the related data.



Lightning Error

Contains details of **errors** that occurred during **user interactions**



Lightning Interaction

Contains details of **user actions** such as clicks, taps, scrolls, etc.



Lightning Page View

Contains details related to **page visit**, **visit duration**, and **load time**



Lightning Performance

Contains details that can be used to **track trends**

Example Event Type: Lightning Page View Event

Whenever a user **loads a page** in Lightning Experience, the event details are stored in a **log file**. Here are some of the **fields and data** that are recorded from a **Lightning Page View** event.

BROWSER_NAME

The name of the **browser** used to access the page

BROWSER_VERSION

The **version** of the browser used to access the page

CLIENT_IP

The **IP address** of the client viewing the application

DEVICE_MODEL

The **name** of the **device** used to view the page (iPhone, etc.)

EFFECTIVE_PAGE_TIME

Time (ms) taken to **load and ready** the page for the user

OS_NAME

The **operating system** of the device used (Android, iOS, etc.)

OS_VERSION

The **version** of the corresponding **operation system**

PAGE_URL

The **relative URL** of the page being viewed by the user

USER_ID

The ID of the **Salesforce user** who performed the request

Custom Component Instrumentation API

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Custom Component Instrumentation API

The Custom Component Instrumentation API can be used to **monitor and track** reported **events or interactions** in custom Lightning web components.

❖ EVENT MONITORING SETTING

To track and monitor activities in Lightning web components in Event Monitoring, the **Enable Lightning Logger Events** in *Setup > Event Monitoring Settings* needs to be **enabled**. The Custom Component Instrumentation API only works for **Lightning web components** and does not support Aura components.

❖ REPORT EVENTS

To log events that can be viewed in Event Monitoring, the custom Lightning web component need to call the **log** function made available via the **lightning/logger module** for Lightning web components.



Custom Component Instrumentation API

A **Lightning web component** can call the **log** function to log data messages to Event Monitoring.

```
<!-- lwcExample.html -->
<template>
  <lightning-button
    label="Update Price"
    onclick={handlePriceUpdate}>
  </lightning-button>
  ...
</template>
```

Calling the `log()` function publishes the data using an **EventLogFile** event type called **Lightning Logger Event** in Event Monitoring.

```
// lwcExample.js
import { LightningElement } from 'lwc';
import updatePrice from '@salesforce/apex/lwcExampleController.updatePrice';
import { log } from 'lightning/logger';
export default class MyComponent extends LightningElement {

  price = 0;
  recordId = '';

  async handlePriceUpdate() {
    await updatePrice({ price: this.price, recordId: this.recordId });
    let message = { type: 'click', action: 'updatePrice' };
    log(message);
  }
}
```

The `log` function is imported from the **lightning/logger** module.

Real-Time Event Monitoring

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Real-Time Event Monitoring

In real-time event monitoring, a captured event can be **streamed** using the publish-subscribe model, and/or it can be **stored** in the database. Here are some other **considerations**.

❖ CONFIGURATION

Real-time events can be configured using the **Event Manager** in Setup, or using the **Metadata API**.

❖ STORAGE

Most real-time event monitoring data are stored in Salesforce **big objects** up to **6 months**. Some events are stored in **standard objects**. The data can be used to **build reports** using Report Builder.

❖ STREAMING

An external application can **subscribe** to the streaming channel using the **EMP** (*Enterprise Messaging Platform*) **Connector**, which is a wrapper built around the **CometD library**.



Real-Time Event Monitoring

In real-time event monitoring, a captured event can be **streamed** using the publish-subscribe model, and/or it can be **stored** in the database. Here are some other **considerations**.

❖ POLICIES

A transaction security can be used to **intercept a real-time event** if the event **matches the conditions** defined in the policy. If so, the action defined in the policy can be executed accordingly.

❖ SUPPORT

It is important to note that **not all** real-time events can be **streamed** as a platform event, **stored** in the database, or **used** in a transaction security policy.



Transaction Security

Transaction Security is a feature that allows Salesforce to respond accordingly to a **captured real-time event** when the event **matches the criteria** of a **transaction security policy**.

❖ ENABLE

Transaction Security is available through **Salesforce Shield** or the **Salesforce Shield Event Monitoring** add-on. To start using, the feature must first be enabled in the **Transaction Security Policies** page in Setup.

❖ POLICY

Once Transaction Security is enabled, an option to create new transaction security policies will be available. A transaction security policy can be created using the **Condition Builder**, or an **Apex class**.



Transaction Security

Transaction Security is a feature that allows Salesforce to respond accordingly to a **captured real-time event** when the event **matches the criteria** of a **transaction security policy**.

❖ ACTIONS

When an event matches a criteria, the triggered transaction security policy can either **allow** or **block** the action, or require the user to **authenticate** using multi-factor authentication (MFA). However, the available options can vary depending on the event type.

❖ NOTIFICATION

A transaction security policy can be configured to **send an email** and/or **in-app notification** to a specified recipient **when** it is triggered.



Creating a Transaction Security Policy

A transaction security policy can be created declaratively using **Condition Builder**, or programmatically using **Apex**.

Create a Transaction Security Policy

Clicks or code?
Choose how you want to define your policy.

Condition Builder



Configure your policy using a point-and-click tool.

Apex



Customize your policy using Apex code.

Next

Transaction Security Policy Conditions

When using the **Condition Builder** to create a transaction security policy, the **Event Type** can be selected from a picklist, then the criteria can be defined by **adding the conditions** as needed.

Create a Transaction Security Policy

What conditions trigger this policy?

Select the event type for this policy, then configure the condition logic. When the conditions are met, the policy triggers and fires a security action. You select an action in the next step.

* Event

Login Event

i

Track login activity. For example, track when users log in from certain locations. [Learn more...](#)

* Condition Logic

All Conditions Are Met (AND)

* Condition

Source IP

* Operator

Does not equal

* Value

XXX.XXX.XXX.XX

+ Add Condition

Back

Next

Transaction Security Policy Action

A policy is triggered when an event **meets its criteria**. Depending on the selected event type, the policy can either **block** or **allow** the event, or require **multi-factor authentication**.

Create a Transaction Security Policy

What actions do you want to take when the policy is triggered?

Choose an action and who gets notified when the policy is triggered. To start enforcing the policy right away, enable the policy.

* Action

Block

Notification ⓘ

☒ Email notification

☒ In-App notification

* Recipient

 Martin Gessner

* Email Notification Content ⓘ

☒ Default Email Content

☐ Custom Email Content

The following default email message will include the OrgID, PolicyID, and UserID, which include the common event data of this action.

* Name

Invalid IP Login

Description

Status ☒ Enabled

Back

Finish

The policy can choose to **block, allow, or require MFA**

An email, in-app notification, or **both** can be sent to a specified Salesforce user.

Real-Time Events

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Real-Time Events

Some of the **real-time events** that can be monitored in an integration solution are described below. These specific events support **storage**, **streaming**, and **transaction security policies**.

Api Anomaly Event

Used to track **anomalies** related to **API calls** performed by users

Api Event

Events related to read-only **API calls** such **query()**, **queryMore()**, and **count()**

File Event

Events captured when a user **previews**, **uploads**, or **downloads** files from the org.

STORAGE



STREAMING



POLICIES



Real-Time Events

Some of the **real-time events** that can be monitored in an integration solution are described below. These specific events support **storage**, **streaming**, and **transaction security policies**.

Login Event

Used to track **login activity** of each user who log in to Salesforce

Session Hijacking Event

When an **unauthorized user** gains access to Salesforce using a **stolen session identifier**

Credential Stuffing Event

When a **user successfully logs** into Salesforce using a credential stuffing **attack**

STORAGE



STREAMING



POLICIES



Example Event: Api Event

When **stored** or **streamed**, this event is referenced as the **ApiEvent** or **ApiEventStream** object respectively. The following describes **some** of the fields recorded from this event.

ApiType

The **API type** that was used in the request (SOAP, REST, etc.)

ConnectedAppId

The ID of the **connected app** associated with the event

ElapsedTime

Time (ms) taken to complete a **query** request

EventUuid

The universally unique **identifier** of the **platform event** message

Operation

The **API call** that was used (Query, QueryAll, or QueryMore)

PolicyId

The ID of the **transaction policy** associated with the event

Query

The **SOQL statement** that was included in the API call

Records

The queried objects' metadata in **JSON string** format

UserId

The ID of the **Salesforce user** who performed the request

Example Event: File Event

While **FileEventStore** events detect file **downloads**, **FileEventStream** events detect file **previews**, **uploads**, and **downloads** from the UI and API. Some of the event fields are described below.

ContentSize

The size (**bytes**) of the file the user interacted with

DocumentId

The unique **18-character ID** of the downloaded file

EventDate

Date and time the activity was detected

EventIdentifier

The **unique ID** of the event

FileAction

Whether the file was **previewed**, **uploaded**, or **download**

FileName

The **name** of the file including its **extension**

FileType

The **content type** of the file

Sourcelp

The **IP address** of the **user** where the request originated

UserId

The ID of the **Salesforce user** who performed the request

Real-Time Events

Some of the **real-time events** that can be monitored in an integration solution are described below. These specific events only support **storage** and **streaming**.

Lightning Uri Event

When a user creates, access, updates, or deletes a record in **Lightning Experience**

STORAGE



STREAMING



Uri Event

When a user creates, access, updates, or deletes a record in **Salesforce Classic**

POLICIES



Example Event: Lightning Uri

When **stored** or **streamed**, this event is referenced as the **LightningUriEvent** or **LightningUriEvent** object respectively. The following describes **some** of the fields recorded from this event.

ConnectionType

The **type** of connection used
(EDGE, GPRS, WIFI, etc.)

DeviceModel

The **name** of the device model
(iPhone, iPad, etc.)

EffectivePageTime

Time (ms) taken to **load and ready** the page for the user

EventUuid

The universally unique **identifier**
of the **platform event** message

Operation

The operation performed such as
read, create, update, or delete

OsName

The **operating system** of the
device used (Android, iOS, etc.)

OsVersion

The **version** of the **operation system**

PageUrl

The **relative URL** of the page
being accessed by the user

RecordId

The ID of the record that is being
viewed or **edited**

Real-Time Events

Some of the **real-time events** that can be monitored in an integration solution are described below. These specific events only support **storage**.

Identity Verification Event

When a user is prompted to **identify** their **identity**

STORAGE



STREAMING



Identity Provider Event

Used to track authentication requests related to **SAML** or **OpenID Connect**

POLICIES



Example Event: Identity Verification Event

This event is referenced as the **IdentityVerificationEvent** object. The following describes **some** of the fields included in the **stored event**.

Activity

The **action** that triggered the **identity verification** request

EventUuid

The universally unique **identifier** of the **platform event** message

Latitude

The latitude of the user based on the user's **IP address**

LoginKey

A string value used to **associate all events** in a user session

Longitude

The longitude of the user based on the user's **IP address**

Policy

The **identity verification** security policy or setting.

Sourcelp

The **IP address** where the request **originated** from

Status

The status of the **identity verification** attempt

VerificationMethod

The method used to **verify identity** in the verification event

Real-Time Events

Some of the **real-time events** that can be monitored in an integration solution are described below.
This specific event only supports **streaming**.

Concurrent Long Running Apex Error Event

When the concurrent long-running Apex limit
has been exceeded

STORAGE



STREAMING



POLICIES



Example Event: Concurrent Long Running Apex Error Event

This event is referenced as the **ConcurLongRunApexErrEvent** object. The following describes **some** of the fields included in the **streamed event**.

CurrentValue

The **number** of concurrent long-running Apex requests

EventUuid

The universally unique **identifier** of the **platform event** message

LimitValue

The **limit value** that was exceeded which triggered the event

Quiddity

The **type of execution** (batch, bulk, future, rest, soap, etc.)

RequestUri

The **URI** of the **page** that received the request

SessionKey

The **session ID** of the user which can be used to **identify events**

Sourcelp

The **IP address** where the request **originated** from

Username

The **Salesforce Username** associated with the request

UserId

The ID of the **Salesforce user** who performed the request

Use Cases

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Scenario



SCENARIO

The Salesforce org of a company has implemented a remote call-in integration solution to allow an external system to perform operations in Salesforce using Bulk API 2.0. The operations may include querying, updating, or deleting thousands of records in Salesforce on a daily basis. The company would like to build a report that would allow them to review any errors or failures that occurred in the batches and see to it that the bulk requests do not go near the limit of batches that can be submitted within a 24-hour period.



Solution



SOLUTION

Event Monitoring tracks activities related to **Bulk API 2.0 requests**. For example, a script can be used to retrieve events from Salesforce where the event type is **BulkApi2** and upload them to a third-party data visualization tool. Using the fields **USER_ID**, **TIMESTAMP**, **OPERATION_TYPE**, **JOB_STATUS**, **RECORDS_FAILED**, and **RECORDS_PROCESSED** contained in each log entry, a report can be established that shows how many batches were submitted by the user in a given day, the status of the jobs as to whether they failed or succeeded as well as the number of records that failed or succeeded in each batch job.



Scenario



SCENARIO

A company has recently adopted Salesforce Connect in their org using an OData 4.0 adapter to enable their sales users to manage orders in an external order fulfillment system from within Salesforce. To gauge user productivity and org stability, department heads would like to monitor the actions that the users perform through Salesforce Connect, status of the performed queries, size of the data sets and the external objects that were accessed, total time consumed in each request, server throughput, and other details.



Solution



SOLUTION

Event Monitoring stores events related to **Salesforce Connect**. Since an OData 4.0 adapter was used, the event type of the related events will be **ExternalODataCallout**. This event type is used to capture information when a user **queries**, **upserts**, or **deletes** records in the external system such as the size (bytes) of the data involved, the associated external object, total time (ms) consumed by the request, and number of records that were retrieved in one second to determine server throughput, and other granular data.



Scenario



SCENARIO

In a Salesforce integration implementation that accepts inbound calls from a remote system using REST API, a solutions architect would like to receive notifications and identify the requests that take up more than 5 seconds to complete. This will allow analysis of the under performing requests and see how the handling can be improved to be able to provide a more efficient service to the remote system.



Solution



SOLUTION

Real-time event monitoring can be used to track events related to **API calls**. In this scenario, storing can be enabled on the **ApiEvent** object so that event gets saved in a big object table in Salesforce which can be queried using batch Apex or Bulk API. To receive near real-time notifications of the event, a **transaction security policy** can be created using the ApiEvent, and a condition is created to match events where the **ElapsedTime** field is greater than 5000 milliseconds. Then, an email or in-app notification can be automatically sent to the solutions architect when an API event is captured that meets the policy conditions.



Learn More

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-  [EventLogFile](#)
-  [EventLogFile Supported Event Types](#)
-  [Introduction to Real-Time Event Monitoring](#)
-  [Real-Time Event Monitoring](#)
-  [Trailhead: Real-Time Event Monitoring](#)
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